

(TR-104) PYTHON WITH AI/ML

Training Day 80 Report

Date: 1 May 2026

Topic Covered: AI Agent Stabilization, Query Validation, Guardrails Enhancement & System Reliability

The eightieth day of training focused on improving AI agent stability, enhancing query validation mechanisms, strengthening guardrails, optimizing search functionality, and increasing overall system reliability through extensive testing and monitoring.

- Resolved recurring Groq API tool execution errors caused by malformed model outputs.
- Implemented a response interception and sanitization layer to ensure valid tool execution.
- Refined system prompts to enforce strict tool usage formats and improve agent reliability.
- Conducted extensive testing of payment, trailer, insurance, customer, and lease-related queries.
- Validated payment status queries, overdue payment retrieval, insurance expiry tracking, and revenue-related analytics.
- Tested multiple query variations to ensure correct intent mapping and tool selection.
- Verified response formatting consistency and system stability across different use cases.
- Performed frontend testing using real-world business queries across customer, trailer, and insurance modules.
- Validated accurate data retrieval and response generation through the frontend interface.
- Improved guardrails by implementing greeting bypass logic for conversational queries.
- Enhanced domain classification and identity-based query handling for more relevant responses.
- Strengthened query validation mechanisms to improve response accuracy and reliability.

- Improved WebSocket logging, connection monitoring, and failure tracking capabilities.
- Optimized search functionality using regex-based and case-insensitive query matching techniques.
- Updated core agent logic, service layers, guardrails, and WebSocket management components.
- Reviewed poster content and provided content for the AI Assistant software poster.
- Revised concepts related to AI model output handling, query validation, frontend testing, guardrails, WebSocket monitoring, and search optimization.

Overall, the training enhanced understanding of AI agent stabilization, query validation techniques, guardrail implementation, frontend testing, system monitoring, and reliable AI-powered application development.